

# National Institute of Technology Tiruchirappalli

## Department of Computer Applications

**Topic Name: Book Recommendation System.**

**Presented By Akanksha Singh**

**MCA**

**under the guidance of**

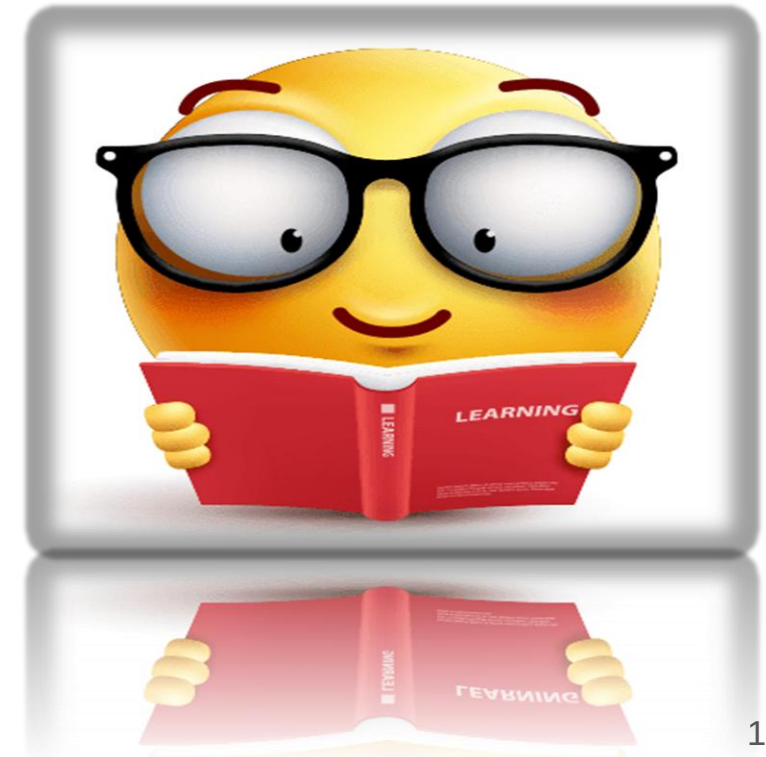
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**NITT**

**04-03-2024**





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# Introduction –Title of Project

*What should I read Next ?*

## ☐ Book Recommendation System



- ☐ The main objective is to create a machine learning model to recommend relevant books to users based on popularity and user interests.



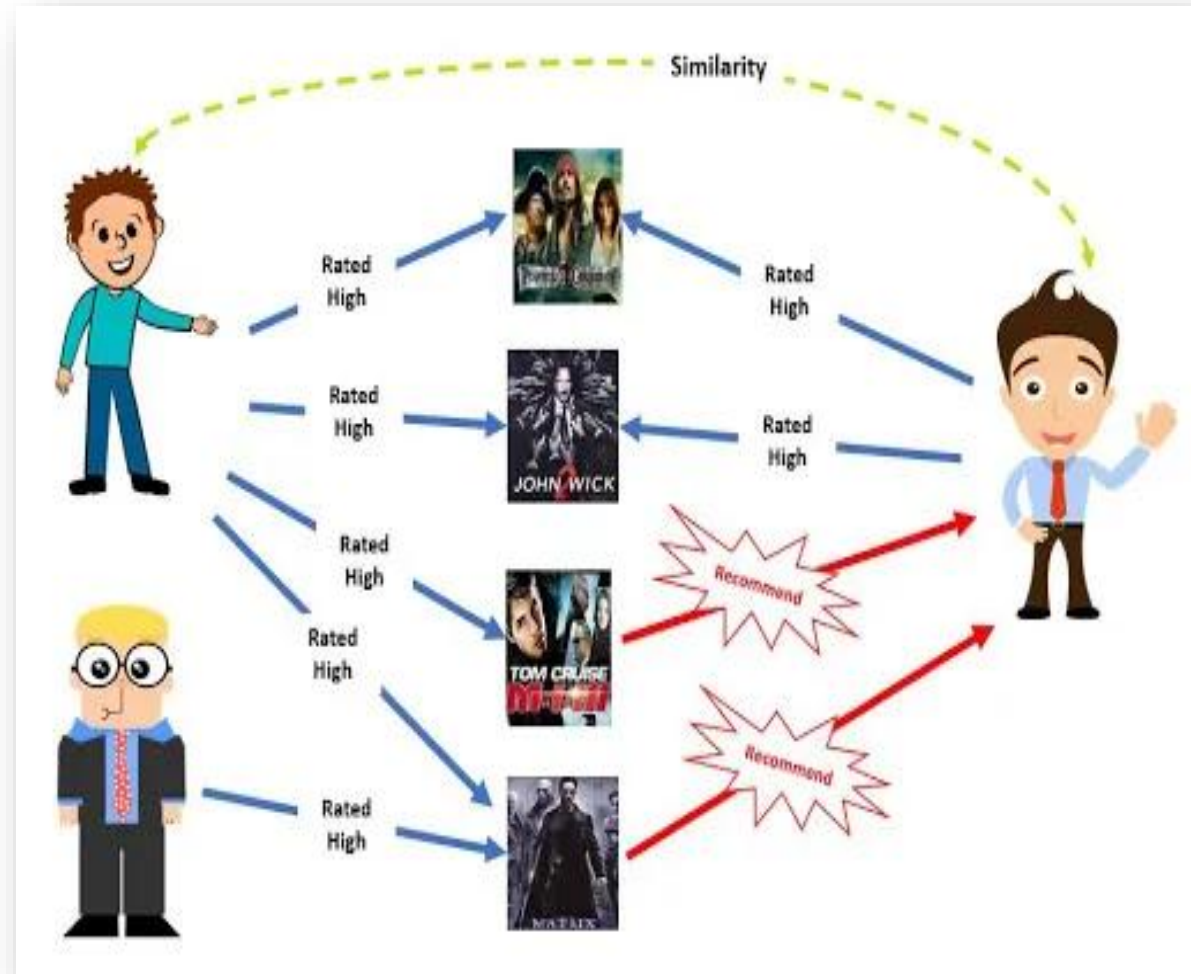
# ***Motivation***

## **WHY?**

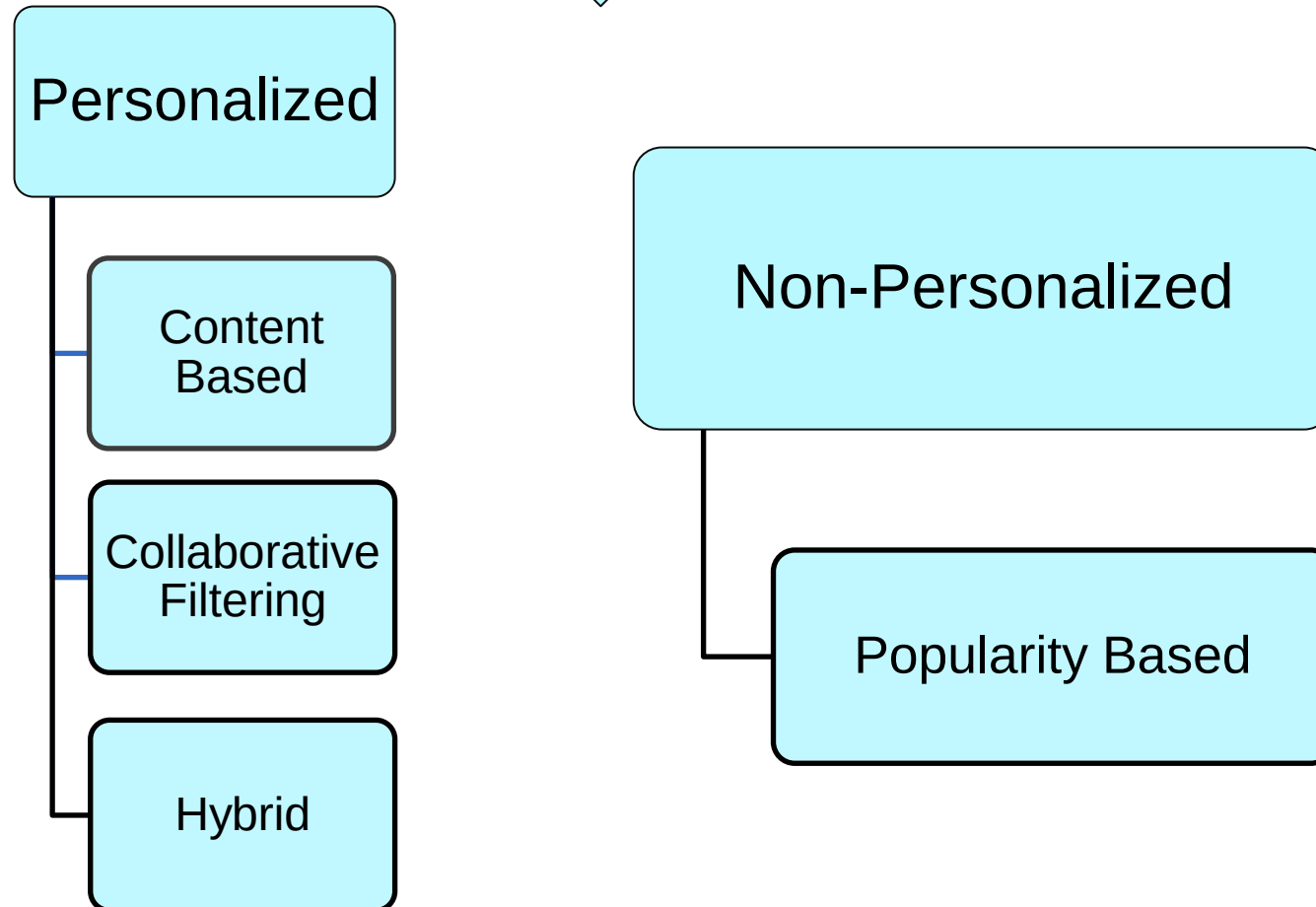
- Benefits users in finding items of their interest.
- Help item providers in delivering their items to the right user.
- Identity products that are most relevant to users.
- Personalized content.
- Help websites to improve user engagement.

# Project Goal

**Explore and compare different approaches to recommending the most relevant books to users based on their interactions (ratings) with other books in the past or based on other similar users' interactions with books.**

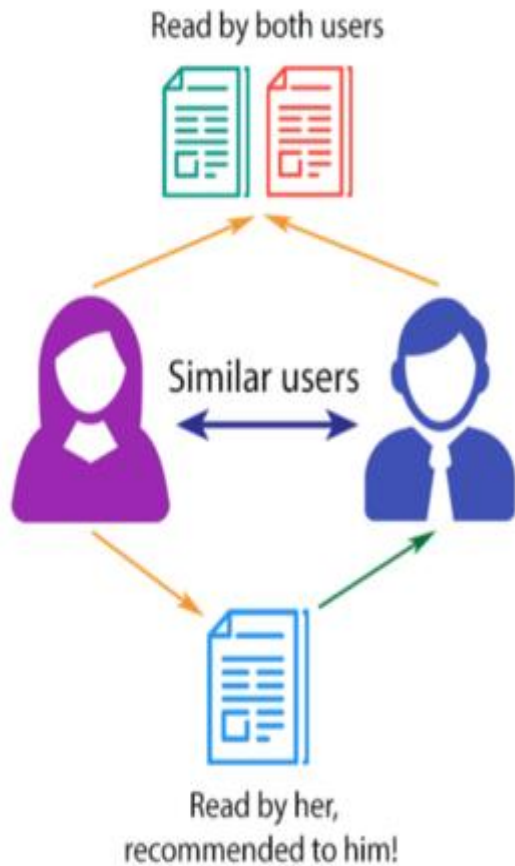


## Recommender System

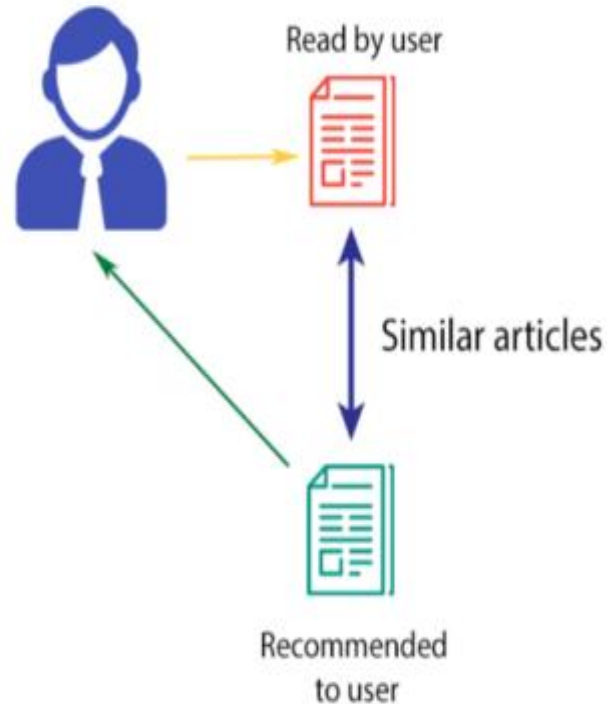


# Types...

## COLLABORATIVE FILTERING



## CONTENT-BASED FILTERING



- Collaborative filtering is achieved using Matrix Factorization.
- **Matrix factorization** is a class of collaborative filtering algorithms used in recommender systems.
- Matrix factorization algorithms work by decomposing the user-item interaction matrix into the product of two lower dimensionality rectangular matrices.

# Data Description

Dataset link:

<https://www.kaggle.com/datasets/arash...>



Three separate files:

Books: Contains basic information regarding the books.

- ISBN, Title, Author, Year of publication, Publisher

Users: Contains basic information regarding the reader.

- User ID, Location, Age

Rating: Contains all of the user rating information.

- User ID, ISBN ,Rating (1 - 10)
- 10 being the highest

# Data Description

```
print(books.head());
```

|   | ISBN       | Book-Title \                                      |  |
|---|------------|---------------------------------------------------|--|
| 0 | 0195153448 | Classical Mythology                               |  |
| 1 | 0002005018 | Clara Callan                                      |  |
| 2 | 0060973129 | Decision in Normandy                              |  |
| 3 | 0374157065 | Flu: The Story of the Great Influenza Pandemic... |  |
| 4 | 0393045218 | The Mummies of Urumchi                            |  |

|   | Book-Author          | Year-Of-Publication | Publisher \             |
|---|----------------------|---------------------|-------------------------|
| 0 | Mark P. O. Morford   | 2002                | Oxford University Press |
| 1 | Richard Bruce Wright | 2001                | HarperFlamingo Canada   |
| 2 | Carlo D'Este         | 1991                | HarperPerennial         |
| 3 | Gina Bari Kolata     | 1999                | Farrar Straus Giroux    |
| 4 | E. J. W. Barber      | 1999                | W. W. Norton & Company  |

|   | Image-URL-S \                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------|
| 0 | <a href="http://images.amazon.com/images/P/0195153448.0...">http://images.amazon.com/images/P/0195153448.0...</a> |
| 1 | <a href="http://images.amazon.com/images/P/0002005018.0...">http://images.amazon.com/images/P/0002005018.0...</a> |
| 2 | <a href="http://images.amazon.com/images/P/0060973129.0...">http://images.amazon.com/images/P/0060973129.0...</a> |
| 3 | <a href="http://images.amazon.com/images/P/0374157065.0...">http://images.amazon.com/images/P/0374157065.0...</a> |
| 4 | <a href="http://images.amazon.com/images/P/0393045218.0...">http://images.amazon.com/images/P/0393045218.0...</a> |

```
3] print(users.head());
```

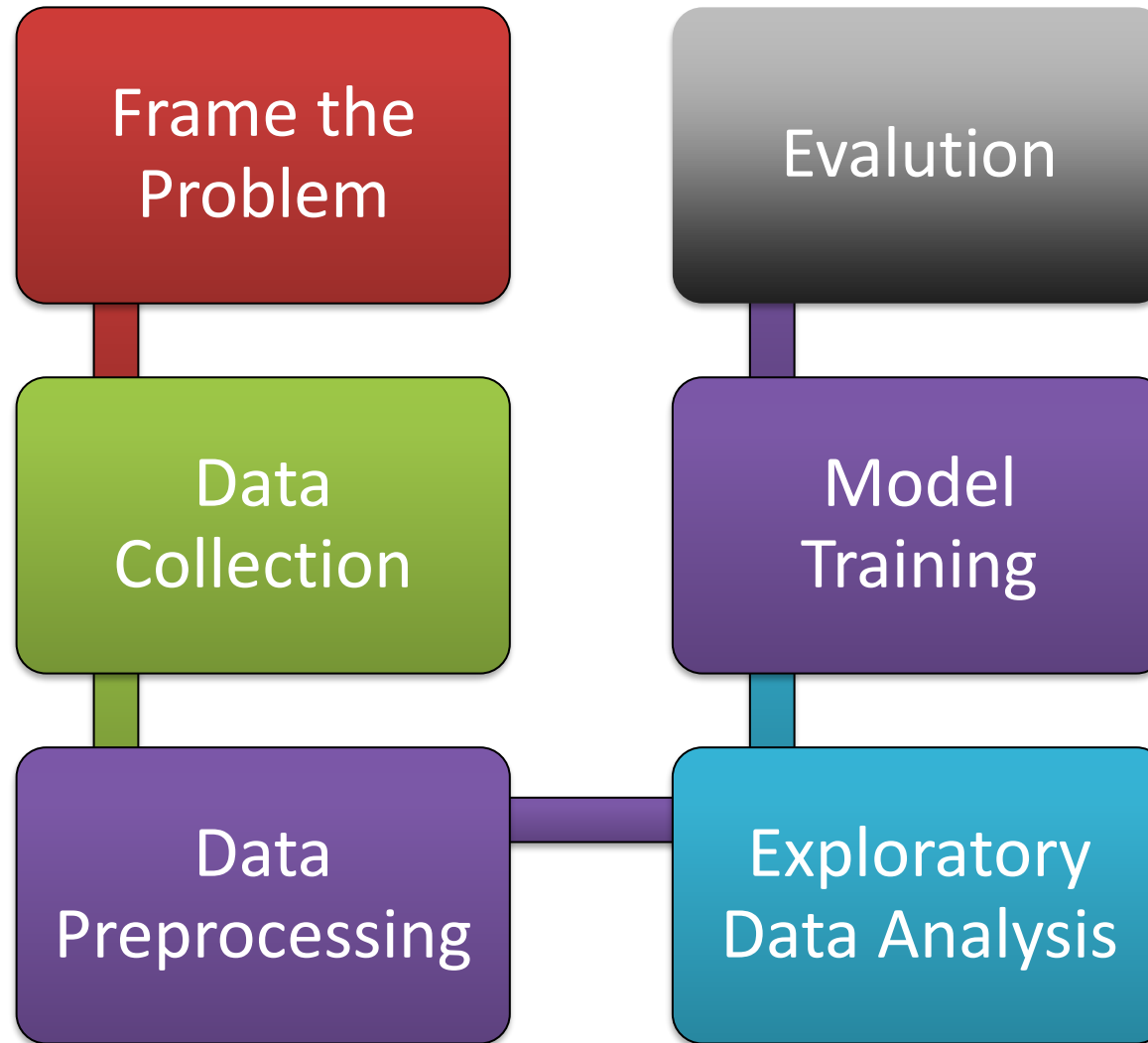
|   | User-ID | Location                           | Age  |
|---|---------|------------------------------------|------|
| 0 | 1       | nyc, new york, usa                 | NaN  |
| 1 | 2       | stockton, california, usa          | 18.0 |
| 2 | 3       | moscow, yukon territory, russia    | NaN  |
| 3 | 4       | porto, v.n.gaia, portugal          | 17.0 |
| 4 | 5       | farnborough, hants, united kingdom | NaN  |

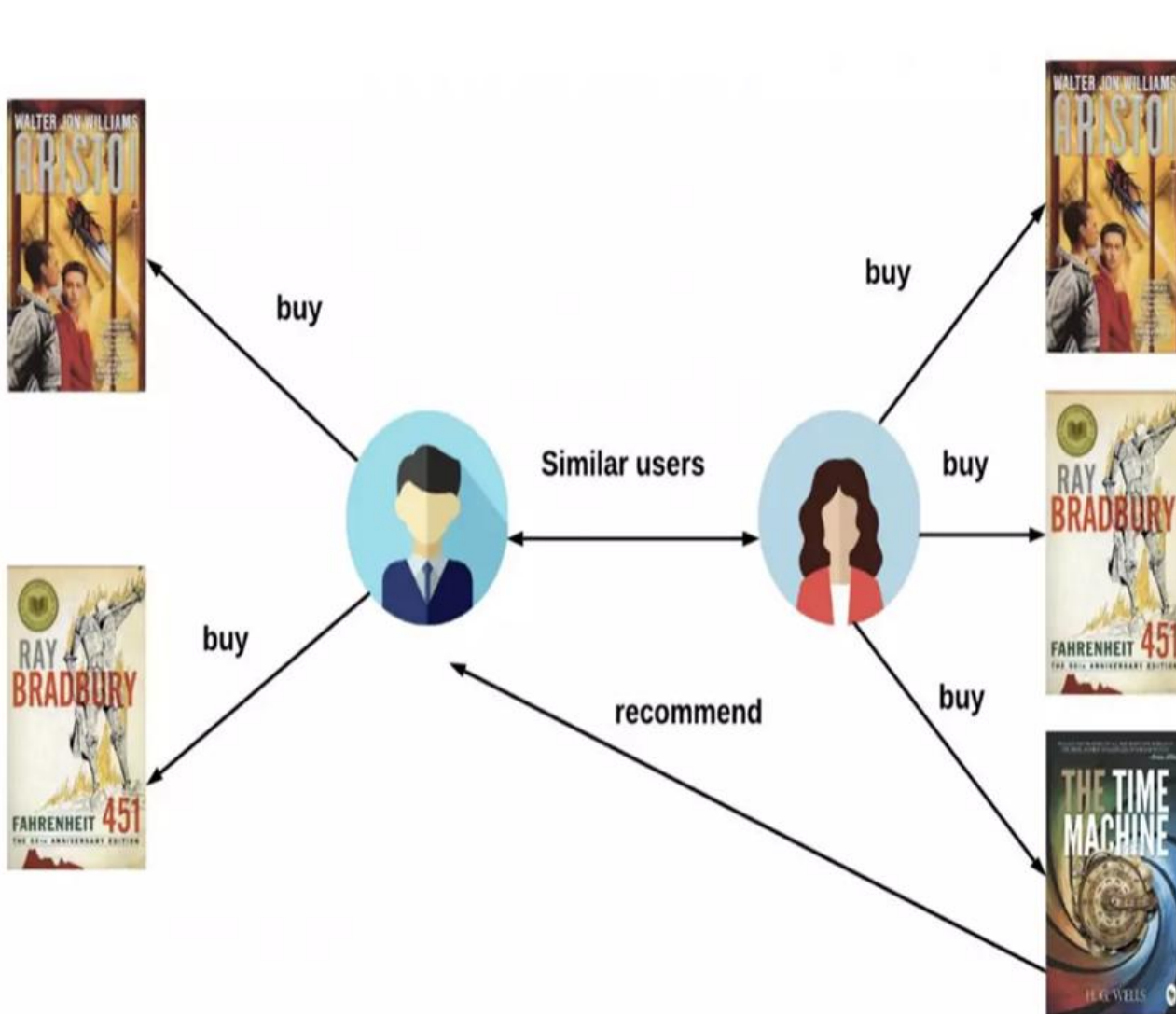
```
print(rating.head());
```

|   | User-ID | ISBN       | Book-Rating |
|---|---------|------------|-------------|
| 0 | 276725  | 034545104X | 0           |
| 1 | 276726  | 0155061224 | 5           |
| 2 | 276727  | 0446520802 | 0           |
| 3 | 276729  | 052165615X | 3           |
| 4 | 276729  | 0521795028 | 6           |



# *Work flow of Project*





**Expected Outcome**



# Data Preprocessing

## In Book Data Set

- ❖ Checking of null values and missing data.
- ❖ Removal of two columns of small image URL and large image URL
- ❖ Changing column names for easy recognition.
- ❖ In the publisher column fill the missing value with others.

## In Users Data Set

- ❖ In the Users dataset in the Age column we find unique values and with that, we calculate the mean age.
- ❖ In the Location columns, we have combined information about the city, state, and country we split this information into three different columns.

## In Ratings Data Set

- ❖ In this data set we check Book-rating and User-Id are columns that are numerical type In the ISBN column we remove extra characters



# Final Data Set Details

After applying preprocessing on all three datasets. We merge all and made the final data set for visualization and model building.

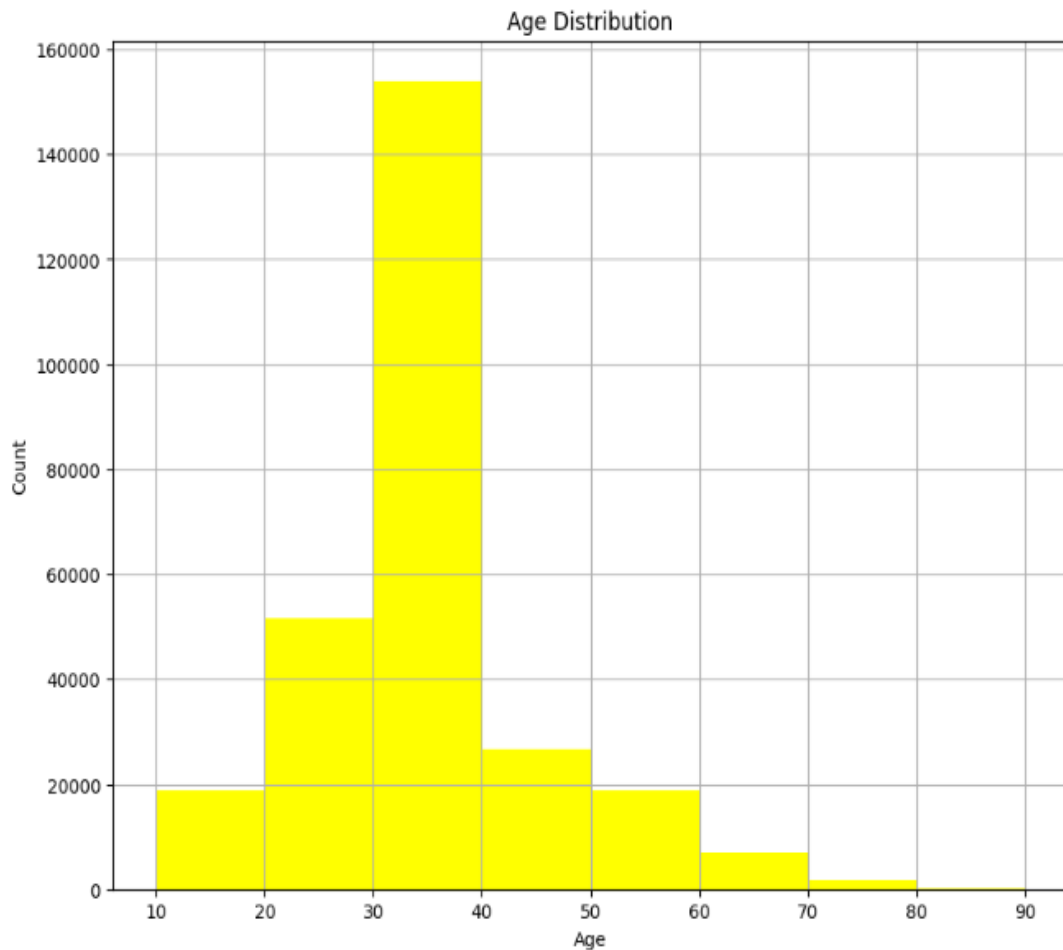
| Datasets                                                                                                                   | Rows And Columns                                                                 | No. Of Unique User-ID                                   | Null Values                                                                                 | Data Types                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Final Dataset</li><li>• After merging of all three preprocessed datasets</li></ul> | <ul style="list-style-type: none"><li>• 50815 Rows</li><li>• 8 Columns</li></ul> | <ul style="list-style-type: none"><li>• 95513</li></ul> | <ul style="list-style-type: none"><li>• Total null values of all 3 datasets are 6</li></ul> | <ul style="list-style-type: none"><li>• Int(32) 1 columns</li><li>• Object 8 columns</li></ul> |

|    | User-ID | ISBN       | Book-Rating | Book-Title   | Book-Author     | Year-Of-Publication | Publisher    | Image-URL-S                                                                                                       |
|----|---------|------------|-------------|--------------|-----------------|---------------------|--------------|-------------------------------------------------------------------------------------------------------------------|
| 63 | 278418  | 0446520802 | 0.0         | The Notebook | Nicholas Sparks | 1996                | Warner Books | <a href="http://images.amazon.com/images/P/0446520802.0...">http://images.amazon.com/images/P/0446520802.0...</a> |
| 65 | 3363    | 0446520802 | 0.0         | The Notebook | Nicholas Sparks | 1996                | Warner Books | <a href="http://images.amazon.com/images/P/0446520802.0...">http://images.amazon.com/images/P/0446520802.0...</a> |
| 66 | 7158    | 0446520802 | 10.0        | The Notebook | Nicholas Sparks | 1996                | Warner Books | <a href="http://images.amazon.com/images/P/0446520802.0...">http://images.amazon.com/images/P/0446520802.0...</a> |
| 69 | 11676   | 0446520802 | 10.0        | The Notebook | Nicholas Sparks | 1996                | Warner Books | <a href="http://images.amazon.com/images/P/0446520802.0...">http://images.amazon.com/images/P/0446520802.0...</a> |
| 74 | 23768   | 0446520802 | 6.0         | The Notebook | Nicholas Sparks | 1996                | Warner Books | <a href="http://images.amazon.com/images/P/0446520802.0...">http://images.amazon.com/images/P/0446520802.0...</a> |

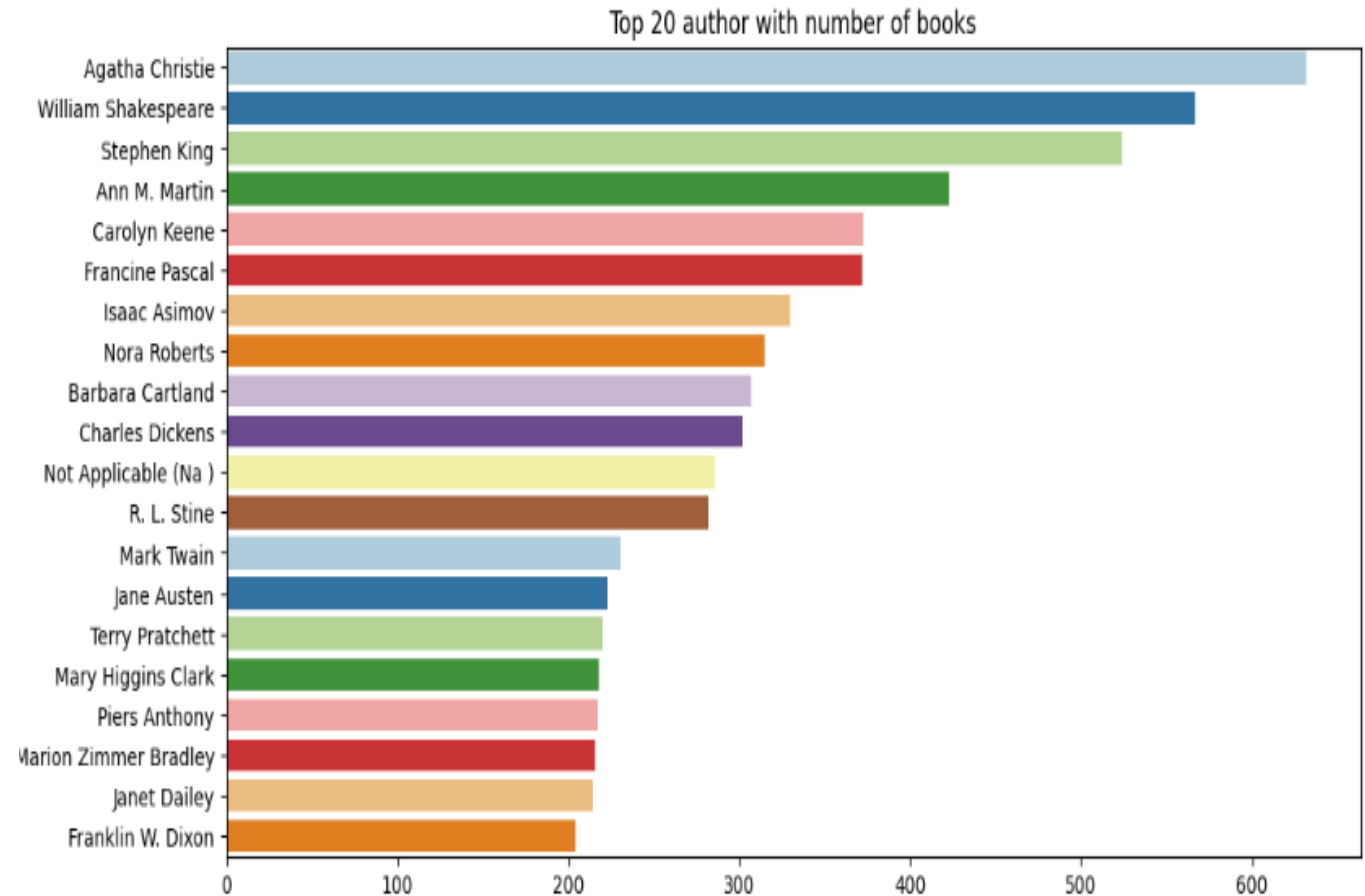


# Visualization

## Analysis no. 1- Age distributions of users data



## Analysis no. 2- Which are the top Author with number of books ?

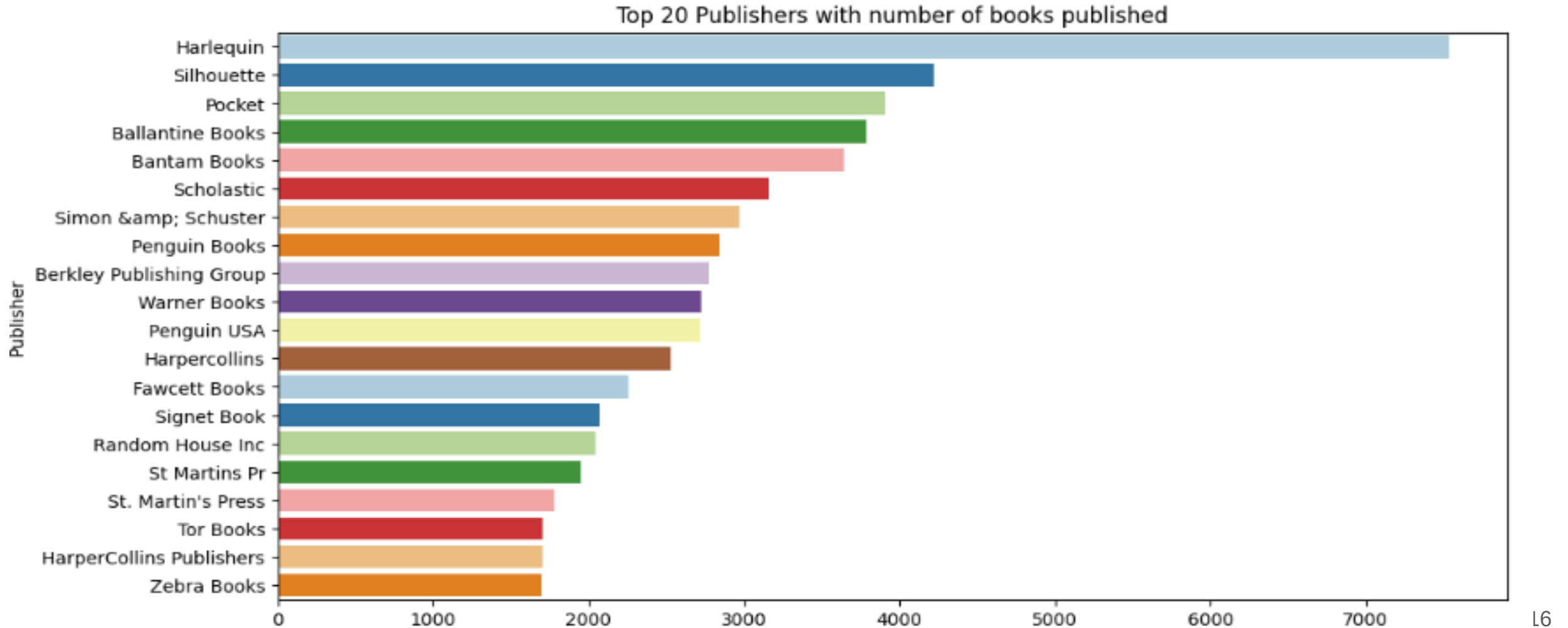




# Visualization

Analysis no.3 -

Which are top publishers with published books ?

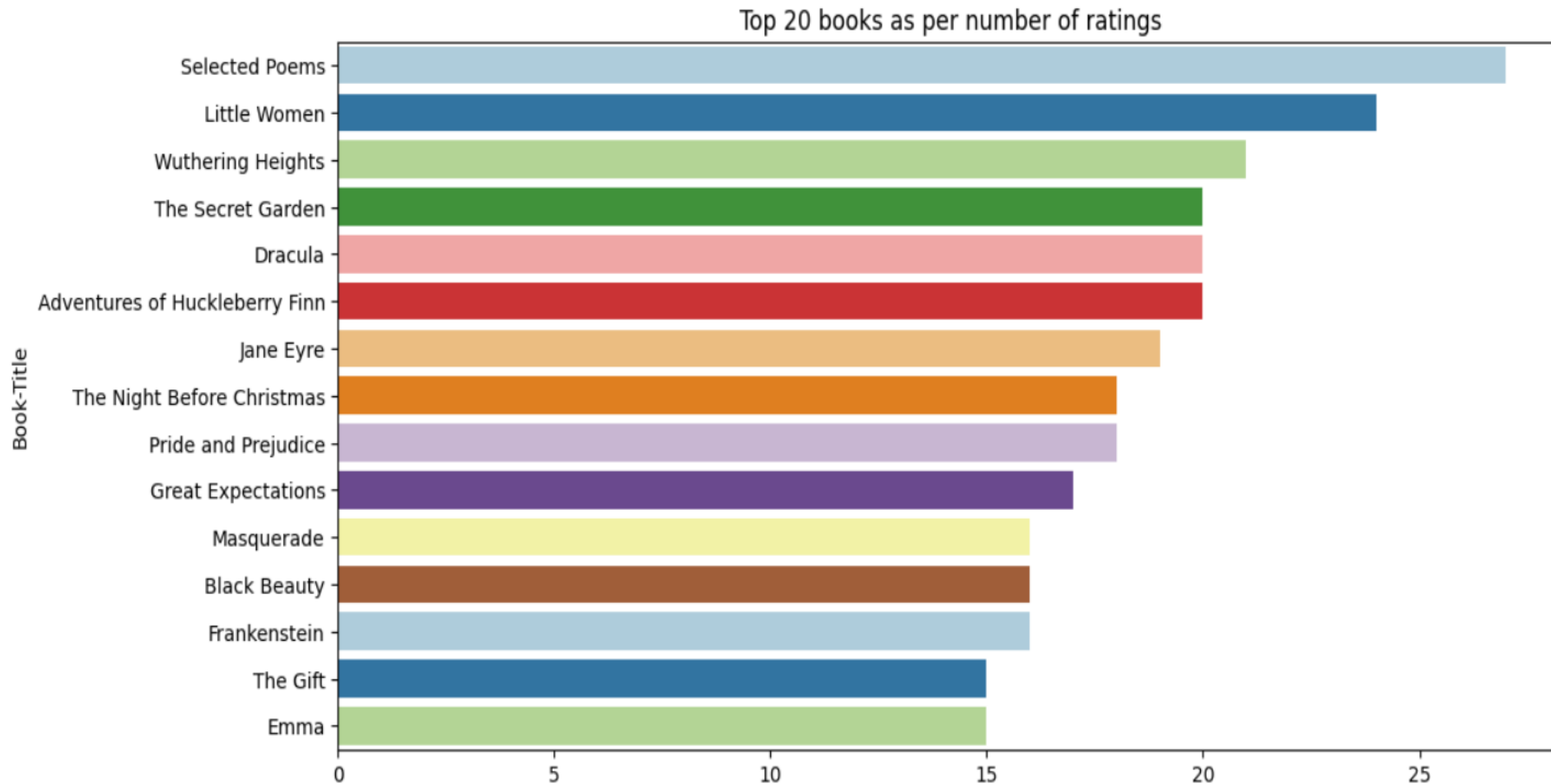




# Visualization

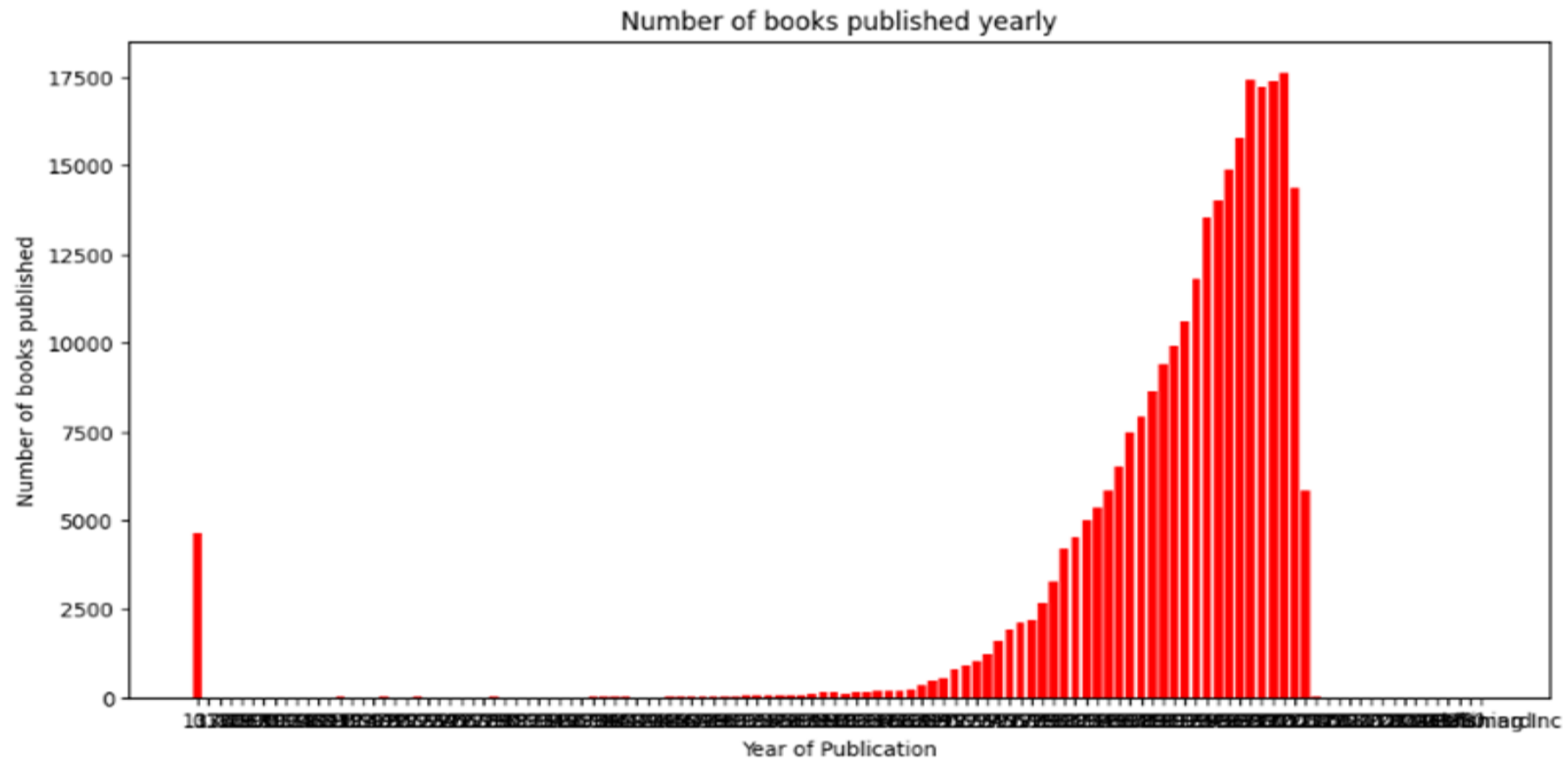
## Analysis No. 4

What are top 20 books as per number of ratings ?

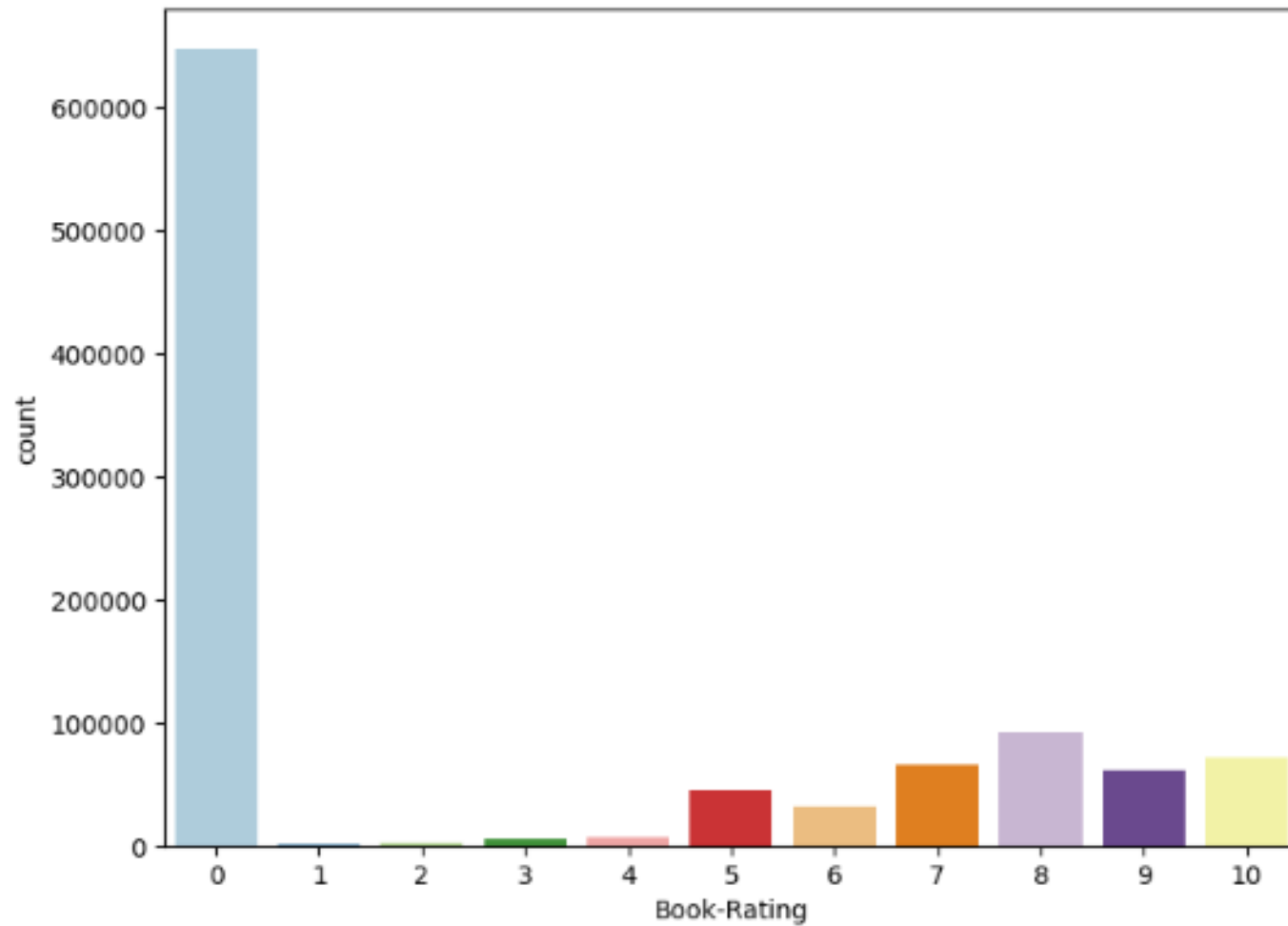


# Visualization

Analysis no. 5-  
Number of Books published in yearly.



## ❑ Analysis On Ratings on Books



❑ This shows users have rated 0 the most, which means they haven't rated books at all.



# Result of Popularity Based System

## ❑ Popularity Based Recommender System

The top ten books as per ratings :

| Book-Title                                      | Book-Rating |
|-------------------------------------------------|-------------|
| Wild Animus                                     | 2502        |
| The Lovely Bones: A Novel                       | 1295        |
| The Da Vinci Code                               | 898         |
| A Painted House                                 | 838         |
| The Nanny Diaries: A Novel                      | 828         |
| Bridget Jones's Diary                           | 815         |
| The Secret Life of Bees                         | 774         |
| Divine Secrets of the Ya-Ya Sisterhood: A Novel | 740         |
| The Red Tent (Bestselling Backlist)             | 723         |
| Angels & Demons                                 | 670         |

### ➤ Advantage of popularity-based recommendation system

- There is no need for the user's historical data

### ➤ Disadvantage popularity-based recommendation system

- This would recommend the same sort of products/books which are scary based on popularity to every other



# Model Selection

## ■ Similarity Based Models

### 1. Cosine Similarity

- Cosine similarity measures the similarity between two vectors of an inner product space.

### 2. K-Nearest Neighbours(KNN) Algorithm

- This approach computes the similarity between users based on their ratings and recommends items liked by similar users.



# Applied Model

## ❑ Cosine Similarity – On Pivot Table

|                     | User-ID | 254 | 2276 | 2766 | 2977 | 3363 | 4017 | 4385 | 6251 | 6323 | 6543 | ... | 271705 | 273979 | 274004 | 274061 | 274301 | 274308 | 275970 | 277427 | 277639 | 278418 |
|---------------------|---------|-----|------|------|------|------|------|------|------|------|------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Book-Title          |         |     |      |      |      |      |      |      |      |      |      |     |        |        |        |        |        |        |        |        |        |        |
| 1984                |         | 9.0 | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | ... | 10.0   | NaN    | NaN    | NaN    | NaN    | NaN    | 0.0    | NaN    | NaN    | NaN    |
| 1st to Die: A Novel |         | NaN | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | 9.0  | ... | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    |
| 2nd Chance          |         | NaN | 10.0 | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | 0.0  | ... | NaN    | NaN    | NaN    | NaN    | NaN    | 0.0    | NaN    | NaN    | 0.0    | NaN    |
| 4 Blondes           |         | NaN | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | 0.0  | NaN  | NaN  | ... | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    |
| A Bend in the Road  |         | 0.0 | NaN  | 7.0  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | NaN  | ... | NaN    | 0.0    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    |



# Result Of Collaborative Based Filtering

▶ `recommend('Message in a Bottle')`

➞ Nights in Rodanthe  
The Mulberry Tree  
A Walk to Remember  
River's End  
Nightmares & Dreamscapes

[190] `recommend('A Walk to Remember')`

The Rescue  
Nights in Rodanthe  
The Notebook  
Granny Dan  
A Bend in the Road

▶ `recommend('Nights in Rodanthe')`

➞ The Mulberry Tree  
Dark Angel  
A Walk to Remember  
GARDEN OF SHADOWS (Dollanger Saga (Paperback))  
The Wedding



# Model Implementation

- In the pivot table, we have lots of zero values and on clustering, this computing power will increase to calculate the distance of zero values so we will convert the pivot table to the sparse matrix and then feed it to the model.
- Now we train our model with **Nearest Neighbors Algorithm**. Which find the distance of every point to every other point.



# Prediction

Let's make a prediction and see whether it is suggesting books or not. we will find the nearest neighbors to the input book id and after that, we will print the top 5 books which are closer to those books.

```
# Example usage:  
recommend_similar_books("2nd Chance", pt, model)
```

```
➤ Top 5 books similar to '2nd Chance':  
1. The Next Accident  
2. Last Man Standing  
3. Exclusive  
4. Four Blind Mice  
5. Unspeakable
```

hence, we have successfully built a book recommendation system.



# References

1. "Retracted: Research on Book Recommendation Algorithm Based on Collaborative Filtering and Interest Degree", *Wireless Communications and Mobile Computing*, vol. 2023, Article ID 9813784, 1 pages, 2023. <https://doi.org/10.1155/2023/9813784>
2. <https://www.geeksforgeeks.org/collaborative-filtering-ml/>
3. <https://www.analyticsvidhya.com/blog/2021/06/build-book-recommendation-system-unsupervised-learning-project/>

Thank You !

Thank You !